

SOCKETS

1.Input:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/socket.h>
#include <sys/un.h>
#include <string.h>

#define SOCKET_PATH "/tmp/my_socket"

int main()
{
    int server_fd, client_fd;
    struct sockaddr_un address;
    char buffer[100];

    // Create socket
    server_fd = socket(AF_UNIX, SOCK_STREAM, 0);

    // Set socket address
    address.sun_family = AF_UNIX;
    strcpy(address.sun_path, SOCKET_PATH);

    // Remove old socket
    unlink(SOCKET_PATH);

    // Bind socket to path
    bind(server_fd, (struct sockaddr *)&address, sizeof(address));

    // Wait for client
    listen(server_fd, 5);

    printf("Server waiting...\n");

    // Accept client
    client_fd = accept(server_fd, NULL, NULL);

    // Receive message
    read(client_fd, buffer, sizeof(buffer));
```

```
client_fd = accept(server_fd, NULL, NULL);

// Receive message
read(client_fd, buffer, sizeof(buffer));

printf("Message from client: %s\n", buffer);

close(client_fd);
close(server_fd);

unlink(SOCKET_PATH);

return 0;
}
```

Output:

```
sushmitha@LAPTOP-B907B06L:~$ touch client.c
sushmitha@LAPTOP-B907B06L:~$ nano client.c
sushmitha@LAPTOP-B907B06L:~$ gcc client.c -o client
sushmitha@LAPTOP-B907B06L:~$
sushmitha@LAPTOP-B907B06L:~$ ./client
sushmitha@LAPTOP-B907B06L:~$
```

2.Input:

```

#include <stdio.h>
#include <unistd.h>
#include <sys/socket.h>
#include <sys/un.h>
#include <string.h>

#define SOCKET_PATH "/tmp/my_socket"

int main()
{
    int client_fd;
    struct sockaddr_un address;
    char message[] = "Hello from client!";

    // Create socket
    client_fd = socket(AF_UNIX, SOCK_STREAM, 0);

    // Set socket address
    address.sun_family = AF_UNIX;
    strcpy(address.sun_path, SOCKET_PATH);

    // Connect to server
    connect(client_fd, (struct sockaddr *)&address, sizeof(address));

    // Send message
    write(client_fd, message, strlen(message) + 1);

    printf("Message sent to server.\n");

    close(client_fd);

    return 0;
}

```

Output:

```

sushmitha@LAPTOP-B907B06L:~$ touch socket.c
sushmitha@LAPTOP-B907B06L:~$ nano socket.c
sushmitha@LAPTOP-B907B06L:~$ gcc socket.c -o socket
sushmitha@LAPTOP-B907B06L:~$ ./socket
Server waiting...
Message from client: Hello Server
sushmitha@LAPTOP-B907B06L:~$ |

```